

NICOLE LUCAS

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EDUCATION

Washington University in St. Louis - St. Louis, MO

May 2025

Bachelor's of Science in Computer Science, Minor: Human-Computer Interaction

Hong Kong University of Science and Technology - Hong Kong

Spring 2024

RELEVANT EXPERIENCE

Pocket Triage - Full Stack Developer & UX Designer | Remote

Dec 2025 – Present

- Engineered an offline-first, multi-language React Native (Expo) application with accessible, low-light-optimized UI components, enabling reliable triage use under intermittent connectivity.
- Architected and extended modular backend services in a typed monorepo (REST APIs, async consumers, communication services) to deliver evidence-based triage guidance via text, voice, and image in ~5 seconds on average.

Equitech Futures - Applied Data Institute Scholar | Remote

Aug 2025 – Dec 2025

- Selected into a research-based, small-cohort data science fellowship focused on data collection, measurement, statistics, and applied AI with real-world impact.
- Applied fellowship learning to the Africa Utility Data Collaborative (AUDC), designing a data analytics platform that surfaces leakage, billing gaps, and compliance risks for water utilities with aging infrastructure.
- Built an interactive Python/Streamlit dashboard that consolidates messy, incomplete operational data into clear KPIs and visualizations, enabling faster, data-informed decisions for non-technical stakeholders.

Classover Inc. - Coding/Robotics Instructor | Queens, NY

Aug 2025 – Present

- Design and deliver weekly Python and robotics curriculum for 20+ students using WhalesBot and Makeblock kits, creating hands-on lessons and projects while maintaining a 95% parent/student satisfaction rate.
- Guided students to build and debug autonomous robots and sensor-driven projects, emphasizing intuitive, real world applications of coding and robotics for everyday problem-solving.

InvestSTL - Data Analyst & UX Researcher | St. Louis, MO

Jun 2023 – Aug 2023

- Developed a Python + SQL pipeline and real-time mapping tool aggregating data for 50+ St. Louis neighborhoods, giving InvestSTL a unified view to guide equity-focused community investments.
- Cleaned and analyzed neighborhood indicators using pandas, Excel, and SQL to uncover trends, contributing to a 25% increase in platform adoption among partner organizations.
- Led UX research sessions with community partners and synthesized feedback into 15+ UX improvements, making complex neighborhood indicators more intuitive and understandable.
- Defined metrics and success criteria amid evolving stakeholder needs to ensure the platform remained aligned with InvestSTL's mission and decision-making workflows.

Washington University Robotics Team - Arm Team Member | St. Louis, MO

Dec 2022 – May 2023

- Programmed a 4-DOF robotic arm with integrated computer vision, enabling autonomous object detection and manipulation with precision control.
- Collaborated with the software team to implement machine learning algorithms, improving servo control accuracy and robotic performance by optimizing motion paths.

Engineering Undergraduate Student Services - Calculus 3 Tutor | St. Louis, MO

Aug 2022 – May 2025

- Led weekly tutoring sessions for engineering undergraduate students, specializing in exam preparation and core calculus concepts.
- Provided individualized academic support, helping 4+ students per semester to strengthen problem-solving skills and boost performance in foundational math courses.

Queens Church - Digital Ministry Volunteer | Queens, NY

Jan 2026 – Present

- Maintain and update the church website and edit Instagram content to improve clarity and community engagement

LEADERSHIP

Society of Hispanic Professional Engineers - President | St. Louis, MO Jan 2023 – May 2025

- Re-established the SHPE chapter post-hiatus, recruiting 100+ members from the ground up and revitalizing campus presence through strategic outreach and programming.
- Managed a \$17.4K annual budget and oversaw planning and logistics for the National Convention, supporting travel, lodging, and registration for 20+ members every year.
- Represented WashU SHPE at the National Institute for Leadership Advancement and National Convention, collaborating with chapter presidents and national leadership to improve programming, increase engagement, and secure mentorships and sponsorships.

WashU Center for Diversity & Inclusion - Student Programming Lead | St. Louis, MO May 2023 – Dec 2024

- Organized and led WashU's inaugural LatinX Summit, engaging 100+ attendees to surface key issues facing Latine students and draft actionable recommendations for university leadership.
- Represented WashU as a student ambassador at high-profile events (including a dinner with Justice Sonia Sotomayor and admissions tabling), strengthening outreach, belonging, and visibility for underrepresented communities.

Alpha Omega Epsilon Beta Xi - Professional Chair Head | St. Louis, MO October 2022 – May 2023

- Planned and executed 10+ workshops, speaker events, and engineering receptions to help further the academic success of 55 members of our chapter.
- Advocated for the establishment of a Diversity, Equity and Inclusion (DEI) board to increase recruitment of underrepresented groups in engineering, resulting in a 25% increase in diverse membership and a more inclusive community.

George and Carol Bauer Leaders Academy - Member | St. Louis, MO Nov 2022 – May 2023

- Participated in the pilot cohort of the George and Carol Bauer Leaders Academy, a purpose-driven leadership development initiative in Washington University in St. Louis focused on building leaders of character and capability.
- Engaged in immersive leadership development experiences including workshops, cohort programs, and worked one-on-one with a leadership coach to set and reach personal and academic goals.
- Completed leadership training emphasizing self-discovery of purpose and values, enhancing self-awareness and interpersonal effectiveness.

HONORS & AWARDS

Annika Rodriguez Scholars Program Issued 2021

A signature scholars program at Washington University in St. Louis that provides a full tuition scholarship. The Annika Rodriguez Scholars Program recognizes students of exceptional merit who demonstrate academic and leadership achievements, as well as the passion for community service.

Pershing Fellowship in Non-Profit Leadership Issued 2023

The Pershing Fellowship provides financial support, programming, and experience to ten Washington University undergraduate students interested in non-profit leadership. Students in the Pershing Fellowship will participate in regular professional development activities, including meeting with leaders from across the St. Louis community, learning about the landscape of non-profit and community organizations in the region, and connecting with cohort and community members.

CERTIFICATIONS

Equitech Futures: Applied Data Institute	Issued 2025
Code Path Course Completion: AI Engineering	Issued 2026
CodePath Course Completion: Advanced Technical Interview Prep	Issued 2026

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C#, JavaScript, HTML, CSS, React, PHP, Swift, AVR Assembly
Tools & Platforms: Git, Docker, Unity, Figma, Netlify, AWS, Azure, SQL, MongoDB, PostgreSQL